Estimating Constrained Functions

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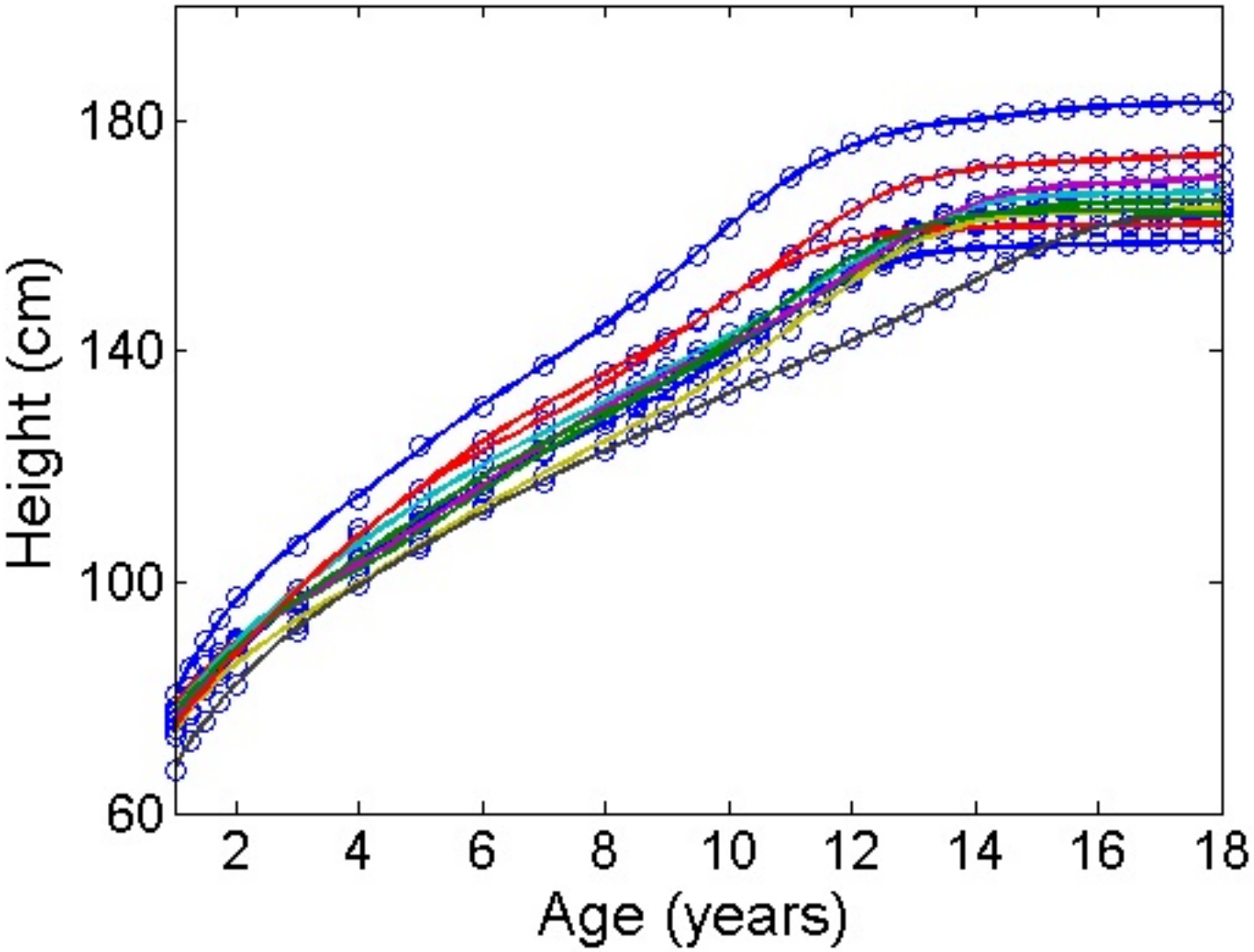
Heights of ten girls



Modeling challenges

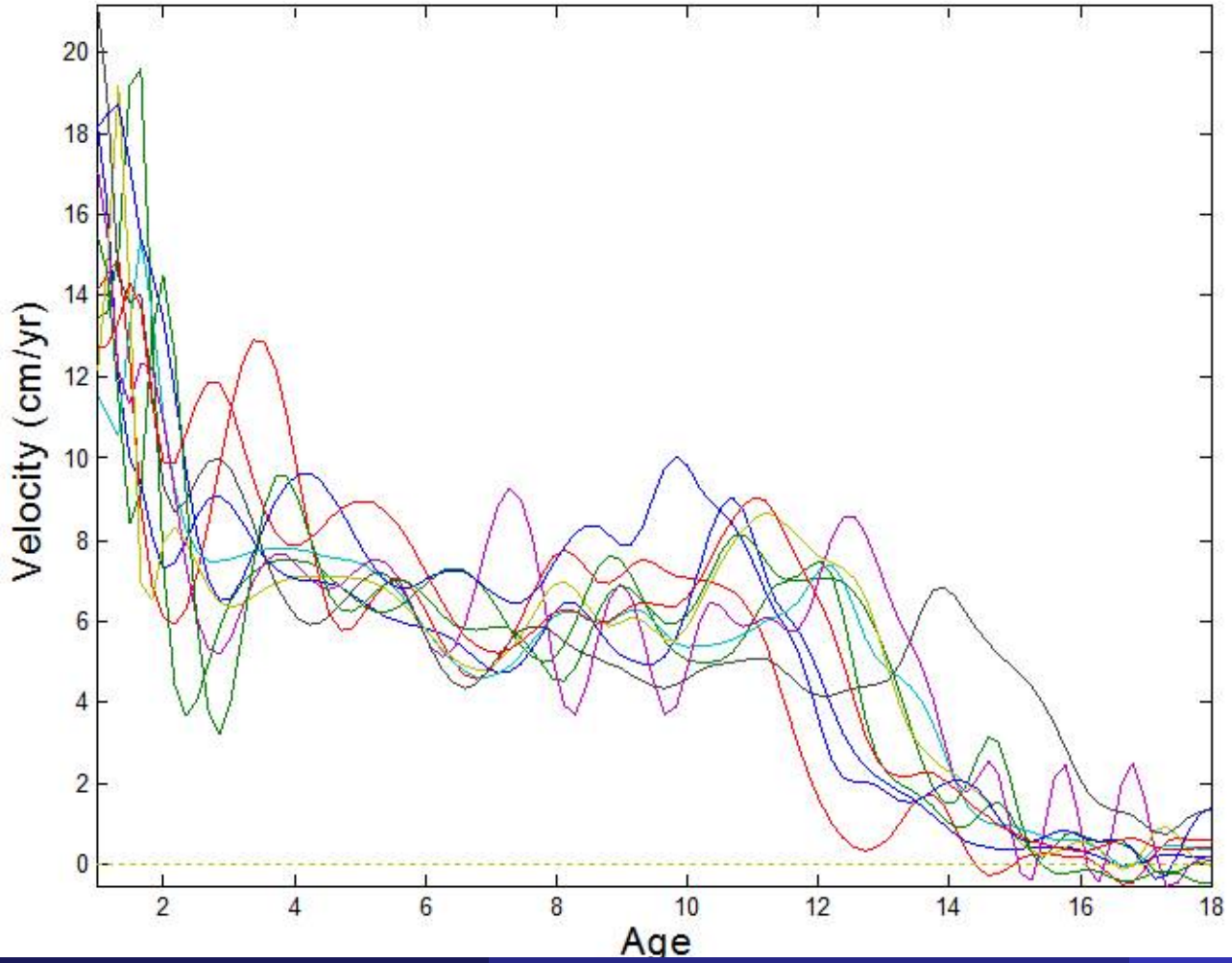
- We want *smooth* curves that fit the data as well as is reasonable.
- We also want to look at velocity and acceleration, so we want to differentiate twice and still be smooth.
- In principle the curves should be *monotone*, that is, have a positive derivative.
- How can we fit a smooth monotone function?

B-splines for growth data

- We use order 6 B-splines because we want to differentiate the result at least twice. Order 4 splines look smooth, but their second derivatives are rough.
- We place a knot at each of the 31 ages.
- The total number of basis functions = order + number of interior knots: 35 in this case.

Let's see what happens?

Noise in the data has a huge impact on derivative estimates.

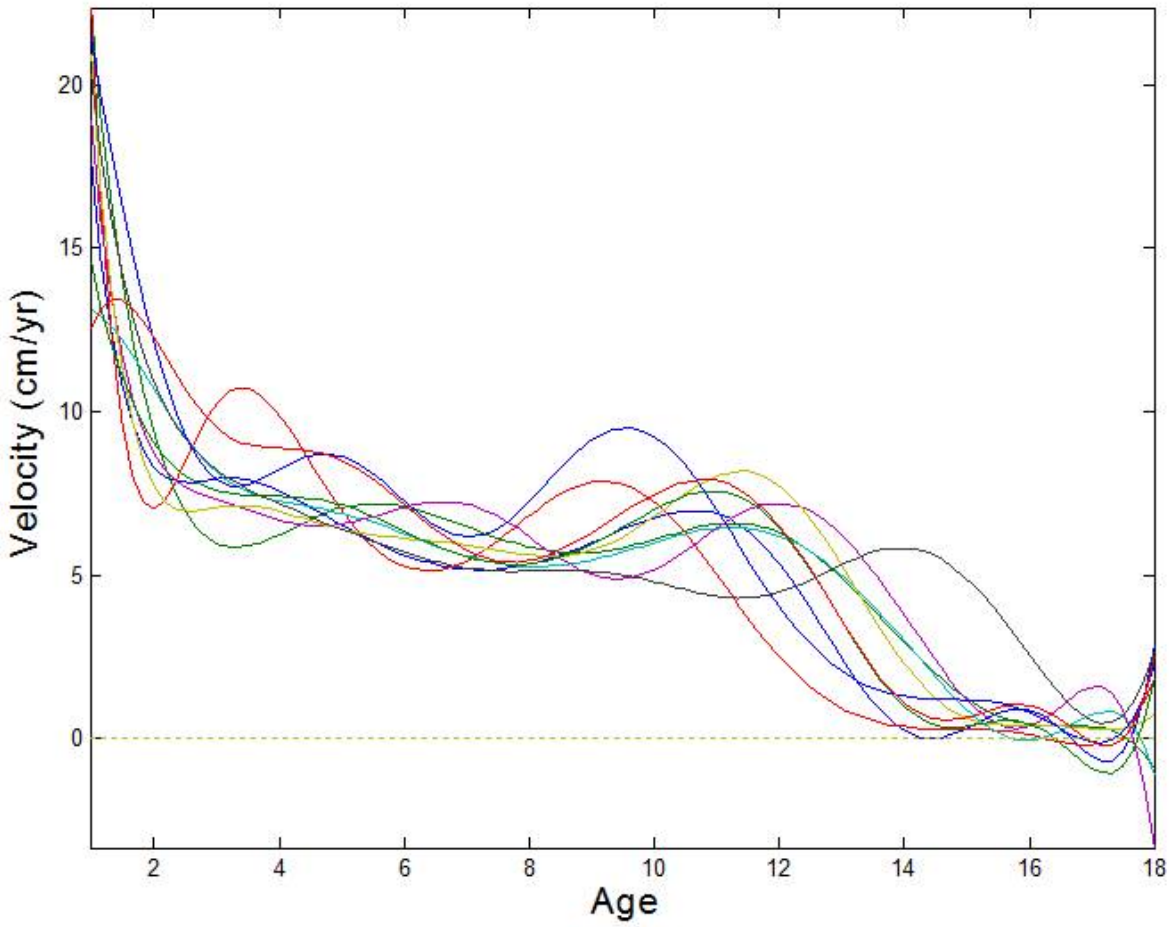


B-splines with fewer knots

Next let's try 12 B-spline basis functions defined by equally-spaced knots.

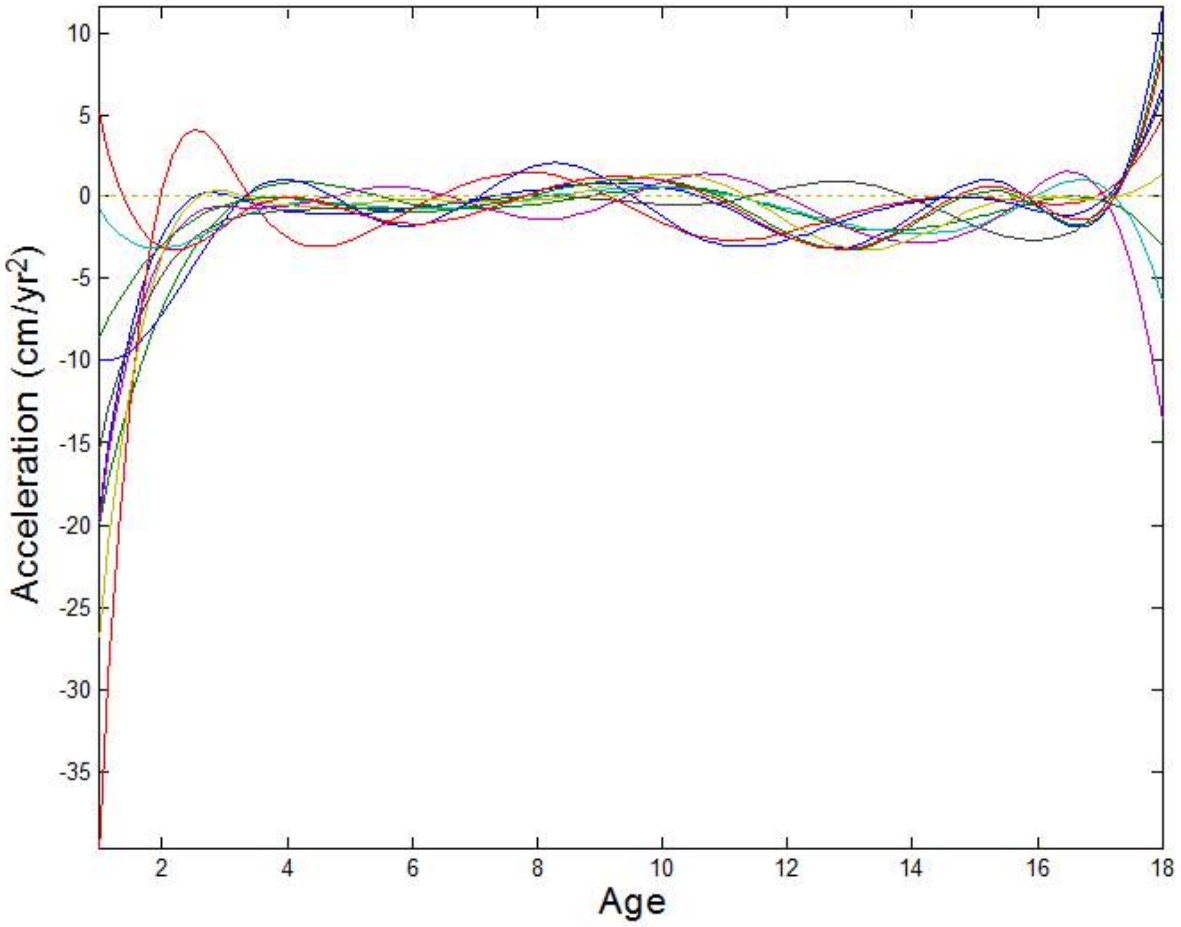
New velocity plot

These velocities are much better. They go negative on the right, though.



Acceleration plot

These acceleration curves are too unstable at the ends. We need something better.



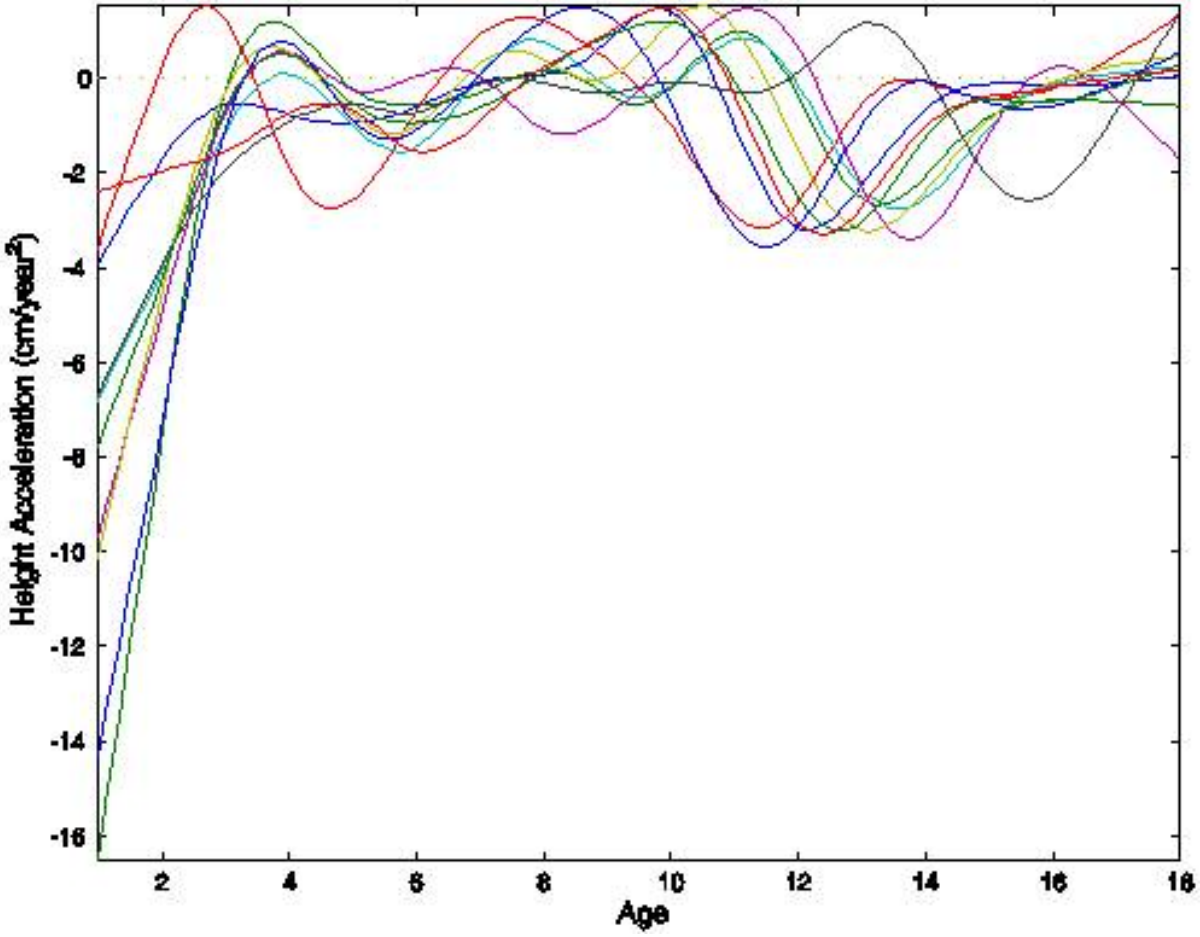
B-splines with roughness penalty

We obtain our fits from minimizing the penalized least squares:

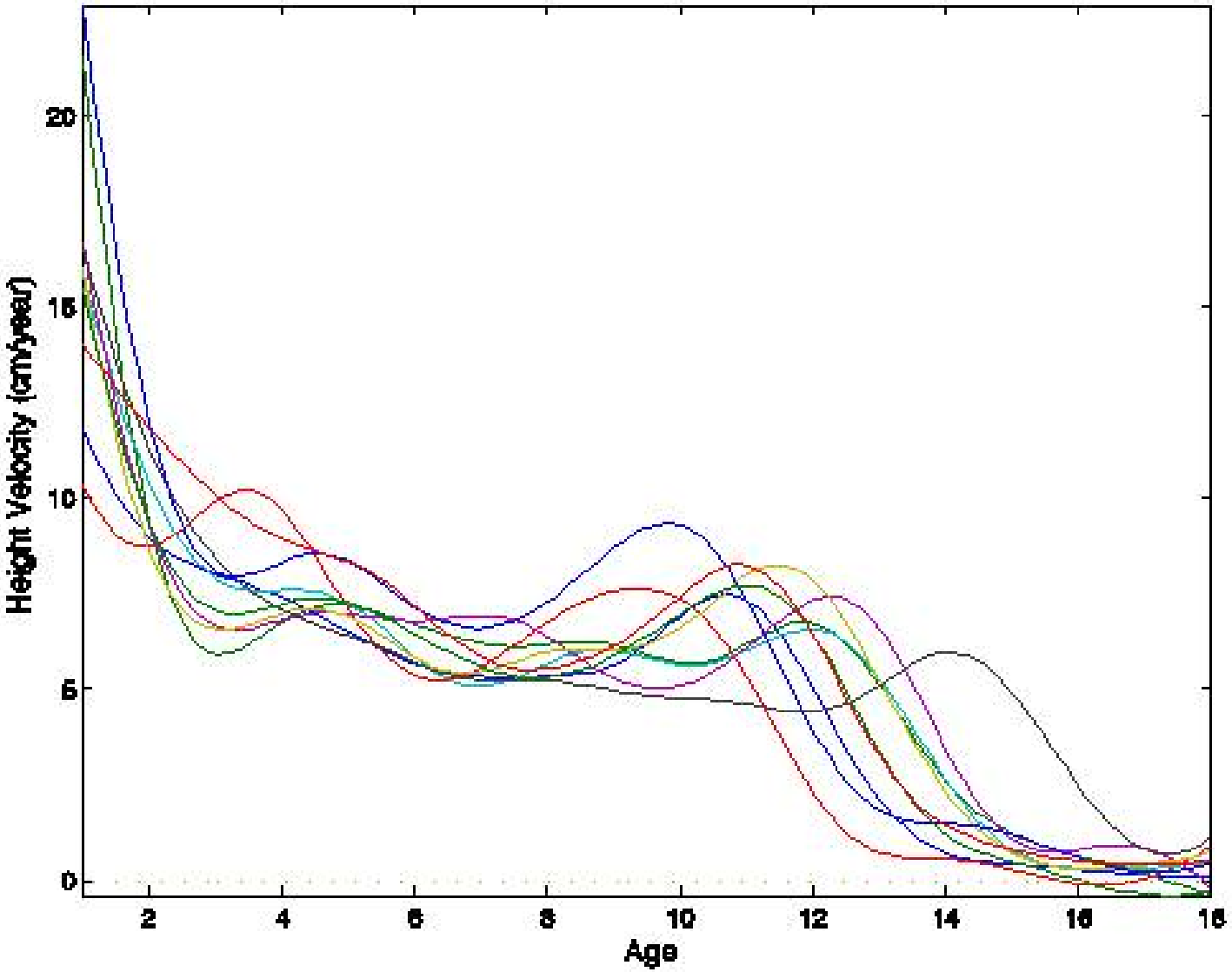
$$\text{PENSSE}_\lambda(x|\mathbf{y}) = [\mathbf{y} - x(\mathbf{t})]'\mathbf{W}[\mathbf{y} - x(\mathbf{t})] + \lambda \int [D^4 x(t)]^2 dt$$

Acceleration plot with roughness penalty

These accelerations are much less variable at the extremes.



Velocity plot with roughness penalty



What about monotonicity?

- The growth curves should be monotonic.
- The velocities should be non-negative.
- Its hard to prevent linear combinations of anything from breaking the rules.
- We need an indirect approach to constructing a monotonic model.

Examples of constraints

- Growth curves should be increasing functions.
- Smooths of magnitudes like precipitation must not be negative.
- Probability density functions have to be positive and integrate to one.
- Functions estimating probabilities have to have values in $[0, 1]$.

Strategy

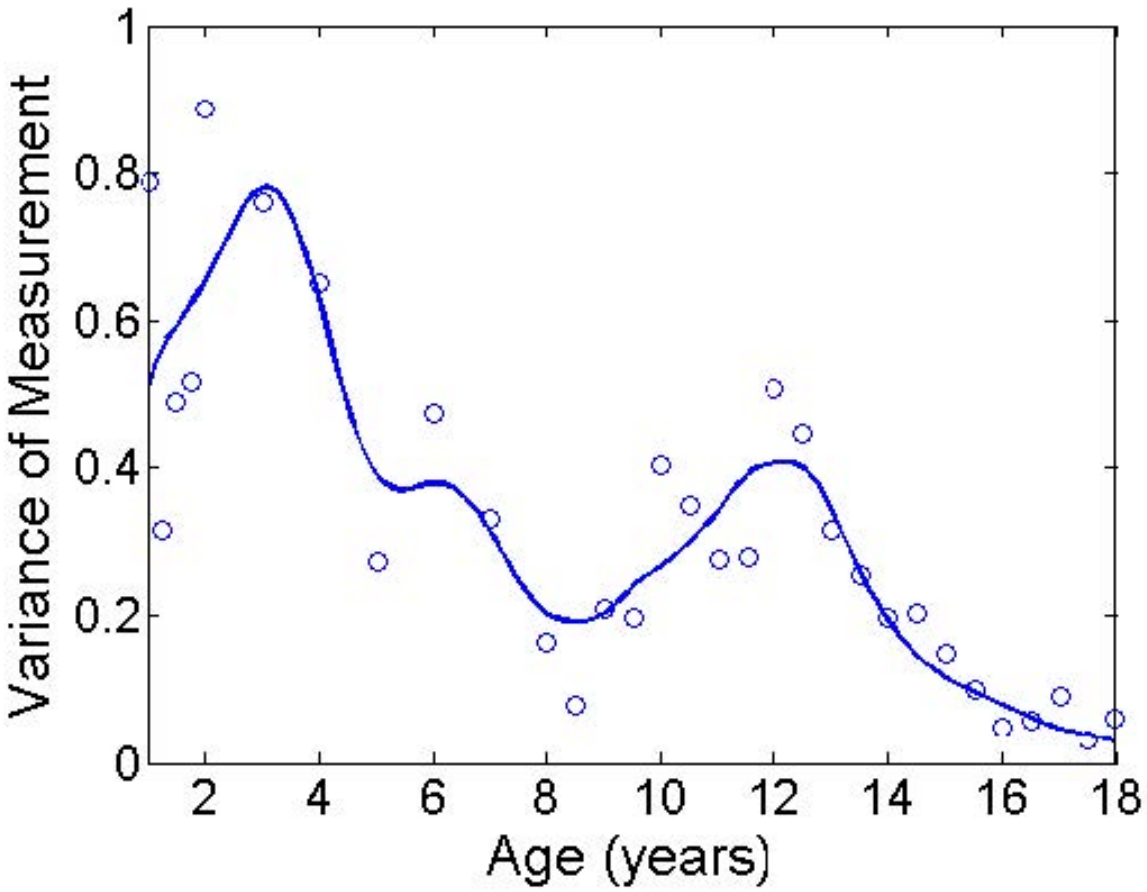
- When curves are constrained in ways such as these, it is not helpful to express them as linear combinations of basis functions since it difficult to enforce the constraints.
- Instead, we express a constrained function as a transformation of an unconstrained function.
- Positive functions can be expressed as exponentials of unconstrained functions, for example.
- Expressing the constrained function in terms of a differential equation can often be helpful.

An example

- The scatter of the observed data about a best-fitting curve is likely to vary over the curve.
- Measurements of height of one year old children have a standard deviation of more than 7 mm, but for adults this is more like 3 mm.
- Consequently we need a curve indicating the variance and standard deviation of errors of measurement.
- But this variance curve must necessarily be positive.

Height variance plot

Circles: squared residuals averaged across the 54 girls in the Berkeley growth study.
Solid line: a positive smooth.



Defining a positive smoothing spline

- We define as the exponential of an unconstrained function W:
$$x(t) = e^{W(t)}.$$
- W is the logarithm of x.
- Since $W(t)$ can be positive or negative and is not in any other way constrained, we can expand W in terms of a set of basis functions,
$$W(t) = \sum_k c_k \phi_k(t).$$
- A Fourier series works for periodic data such as precipitation and B-spline serve for non-periodic data such as the mean squared residuals for the growth data.

Penalizing roughness

- The roughness of positive smoothing function x is determined by the roughness of its logarithm W.
- For example, we may fit the data by minimizing

$$[y - e^{W(t)}]'\mathbf{W}[y - e^{W(t)}] + \lambda \int [D^2 W(t)]^2 dt.$$

- We must resort to numerical optimization methods to minimize this or some other fitting criterion with respect to coefficients c_k.
- The growth variance curve was estimated using order 4 B-splines with knots at data locations and $\lambda = 0.0001$.

Defining a positive function by differential equation

- The transformation $x(t) = e^{W(t)}$ can be expressed as

$$Dx(t) = w(t)x(t).$$

where

$$w(t) = DW(t).$$

- Positive function x is the solution of a simple *differential equation*.
- This definition of x may seem pedantic and un-intuitive, but we will see that it is a powerful tool for defining other constrained functions.

An example

- Growth curves should not in principle lose height, even when the noisy data themselves don't always increase.
- It is difficult to prevent smoothing splines from doing this, especially near adulthood, when the curves flatten.
- Daily measurements of the length of the tibia of a newborn baby, accurate to within about 0.1 mm, challenge our capacity to keep curve monotone.

Defining a monotone smoothing spline

- We express $Dx(t)$ as the exponential of an unconstrained function W:

$$Dx(t) = e^{W(t)}.$$

- $$x(t) = C + \int_{t_0}^{t} \exp[W(u)]du,$$

  where C is a constant to be estimated.

Defining a monotone function by differential equation

- The transformation $Dx(t) = e^{W(t)}$ can be expressed as
$$D^2x(t) = w(t)Dx(t).$$
Here $w = D^2x/Dx$ is the derivative of the logarithm of velocity.
- General solution to the differential equation:
$$x(t) = C_0 + C_1 \int_{t_0}^{t} \exp[\int_{t_0}^{u} w(v)dv]du.$$

Role of w

For example,

- when $w(t) = 0$ for all t, then

$$x(t) = C_0 + C_1 t.$$

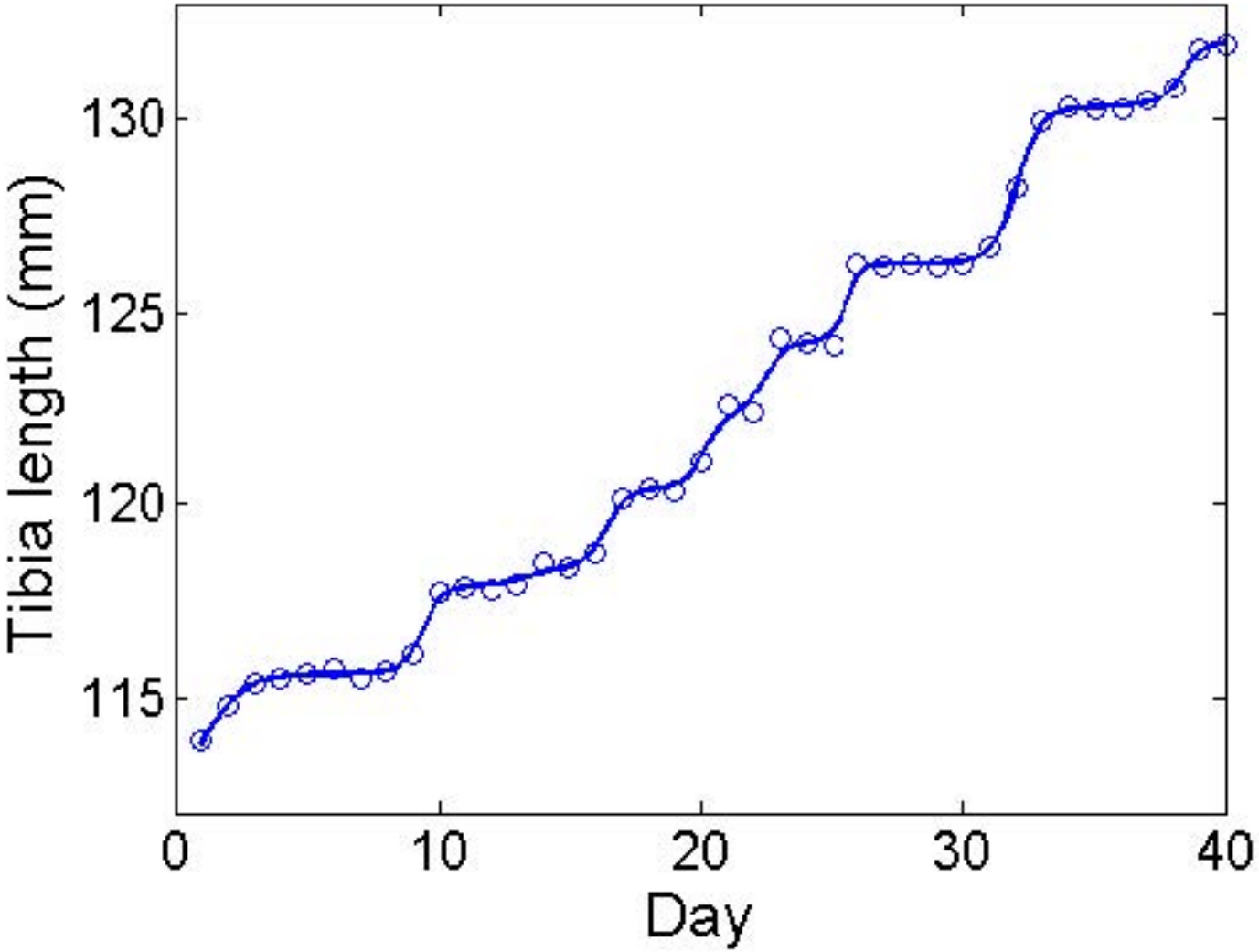
- when w is a nonzero constant,

$$x(t) = C_0 + C_1 e^{wt}.$$

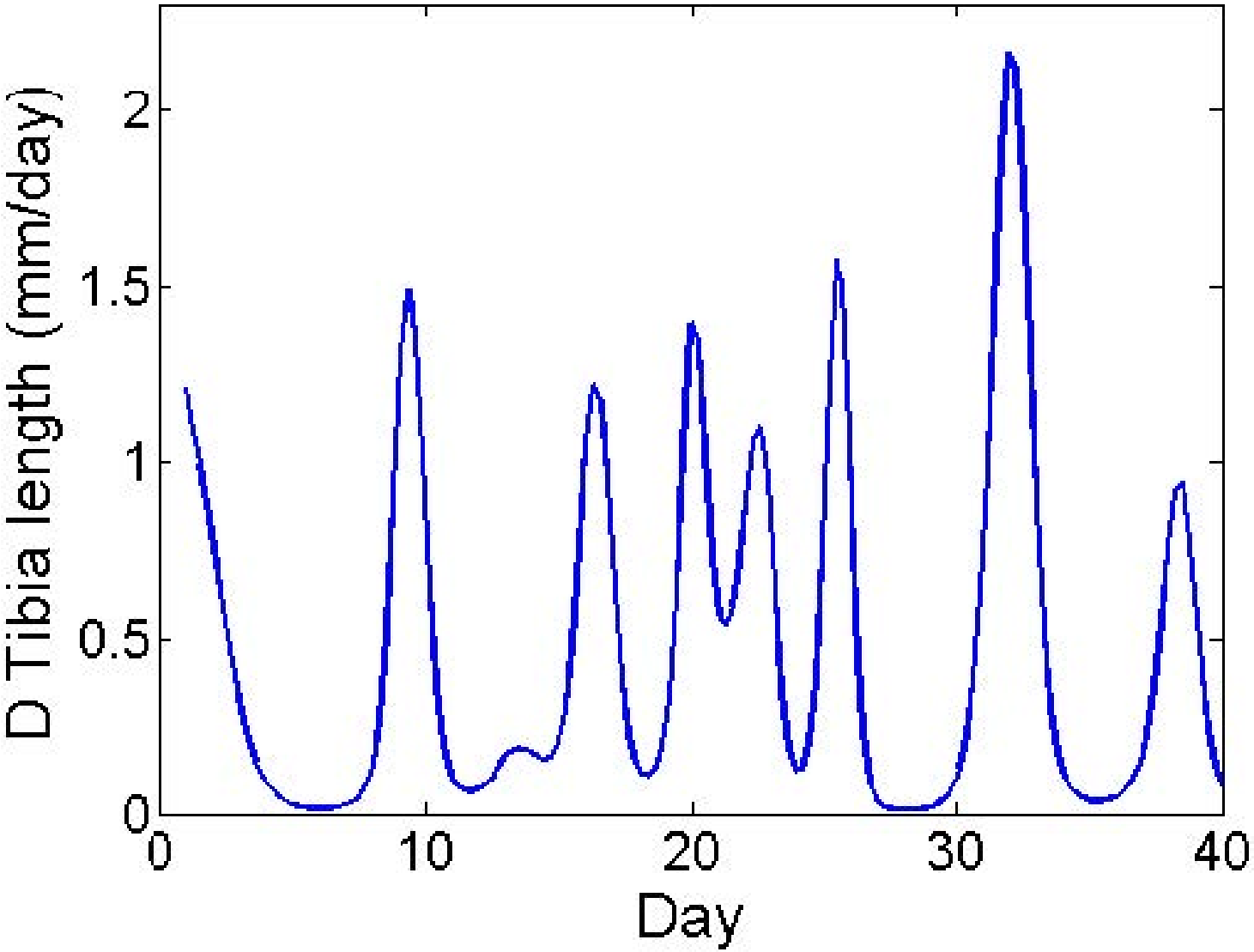
- If w is a function, the closer to zero it is at a value t, the more its local behavior around t will be linear.
- If $C_1 > 0$, then positive values of $w(t)$ imply local exponential increase, and negative values of $w(t)$ imply locally approaching some constant in an exponential rate.

Height of a newborn baby

Solid line: a monotone smoothing spline fit.



Velocity of the baby's growth



Why density estimation?

- Most approaches to smoothing either implicitly or explicitly assume that the scatter of the data about the curve has a normal or Gaussian distribution, and usually with a constant standard deviation.
- But we often see decidedly non-Gaussian behavior in the actual scatter:
  - long tails
  - pronounced skewness
  - multi-modality
- We also see the nature of the scatter varying over space, time, and other variables.

Why density estimation?

- We need the technology to smooth data and estimate the density function for the scatter (i.e., residuals) at the same time.
- The density function may need to be a density surface, varying in shape smoothly over t.

Defining smoothing splines for density estimation

- We express a probability density function $p(t)$ as:

$$p(t) = e^{W(t)} \Big/ \int e^{W(s)} ds \, .$$

- The log likelihood is

$$\sum_{i=1}^{N} \log p(t_i) = \sum_{i=1}^{N} W(t_i) - N \log \int e^{W(s)} ds.$$

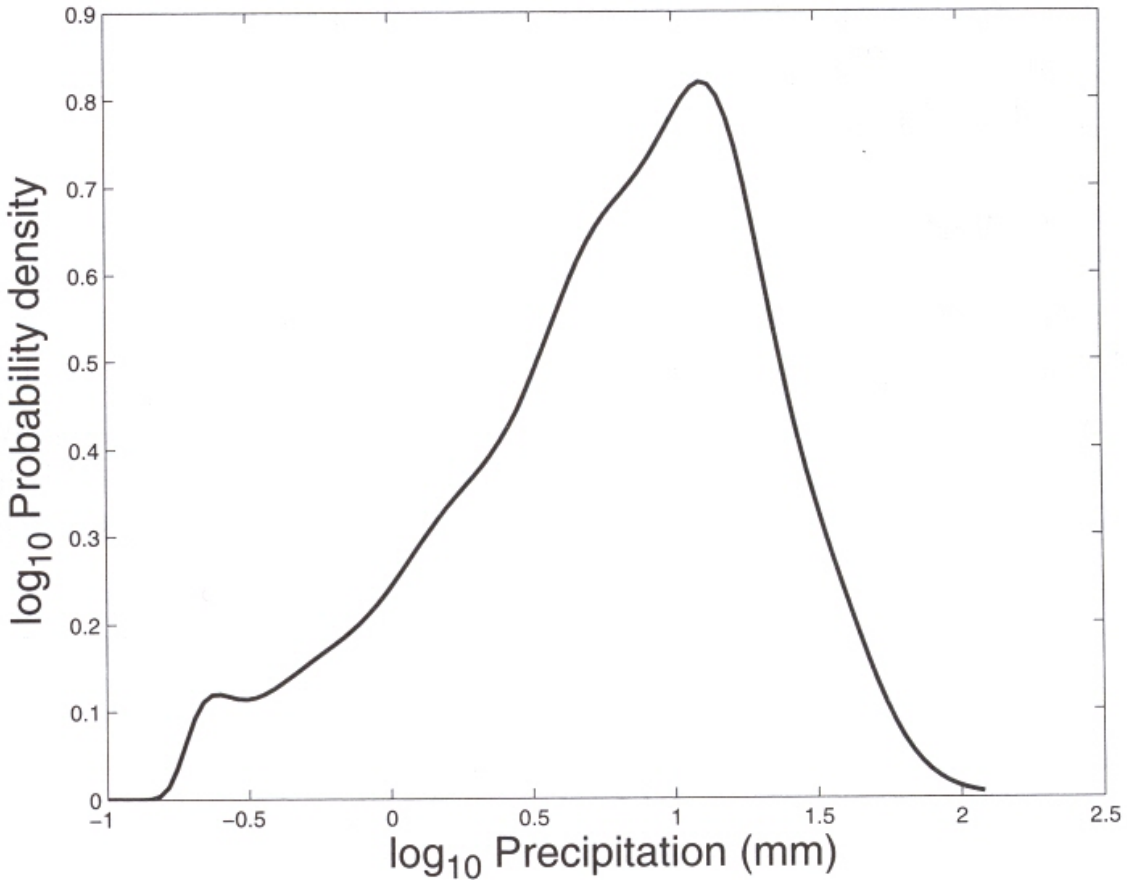
where C is a constant to be estimated.

- Maximizing the penalized log likelihood with an appropriate roughness penalty.

Example: Precipitation at Prince Rupert, Canada

- To show the p.d.f. for the log of daily rainfall at Prince Rupert, British Columbia, one of the rainiest places in North America, over the years 1960 through 1994.
- About a third of the days have a precipitation of 0.1 mm or less, so we used only the 7697 days having precipitation in excess of 0.1 mm.
- Sixteen B-spline basis functions of order 5 and equally-spaced knots were used to expand $W(t)$, and the size of the 3rd derivative was penalized with $\lambda = 10^{-6}$.

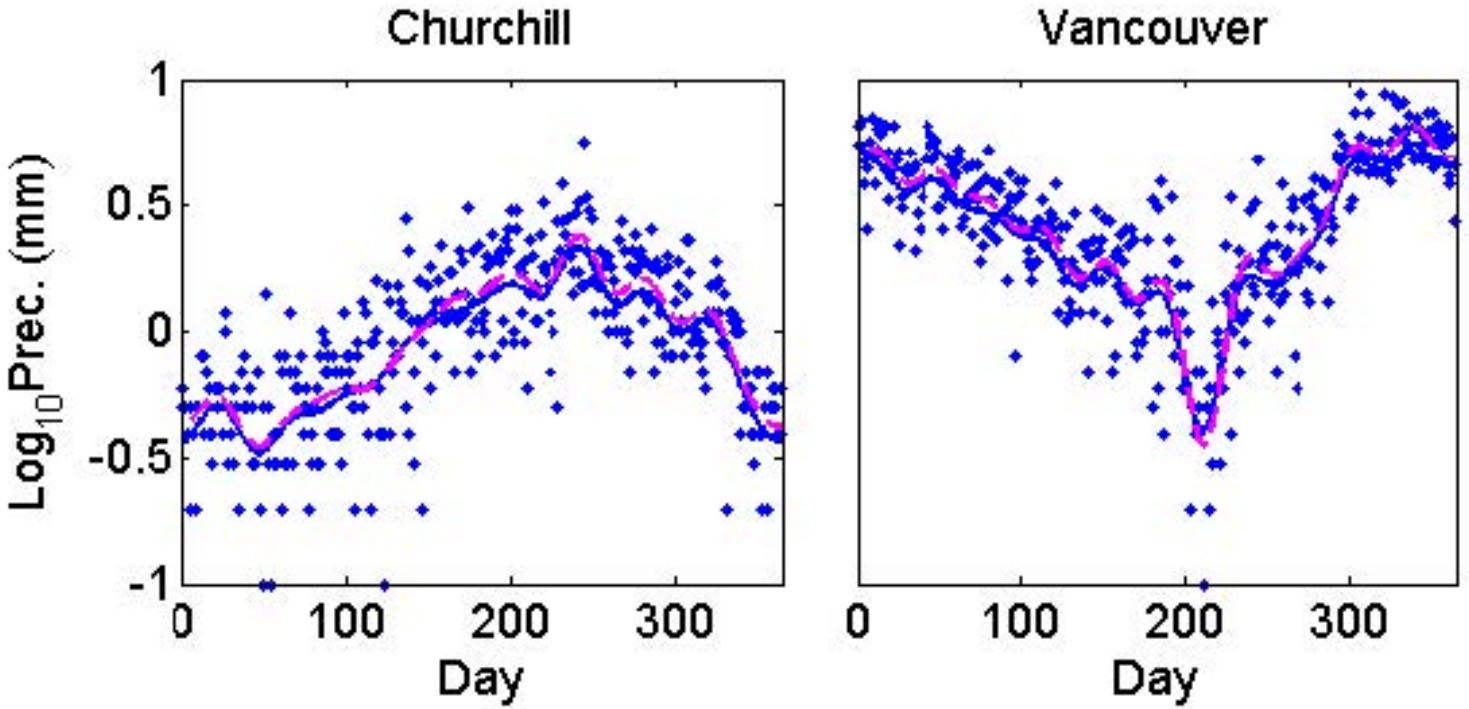
Estimated PDF for Log10 Daily Precipitation at Prince Rupert



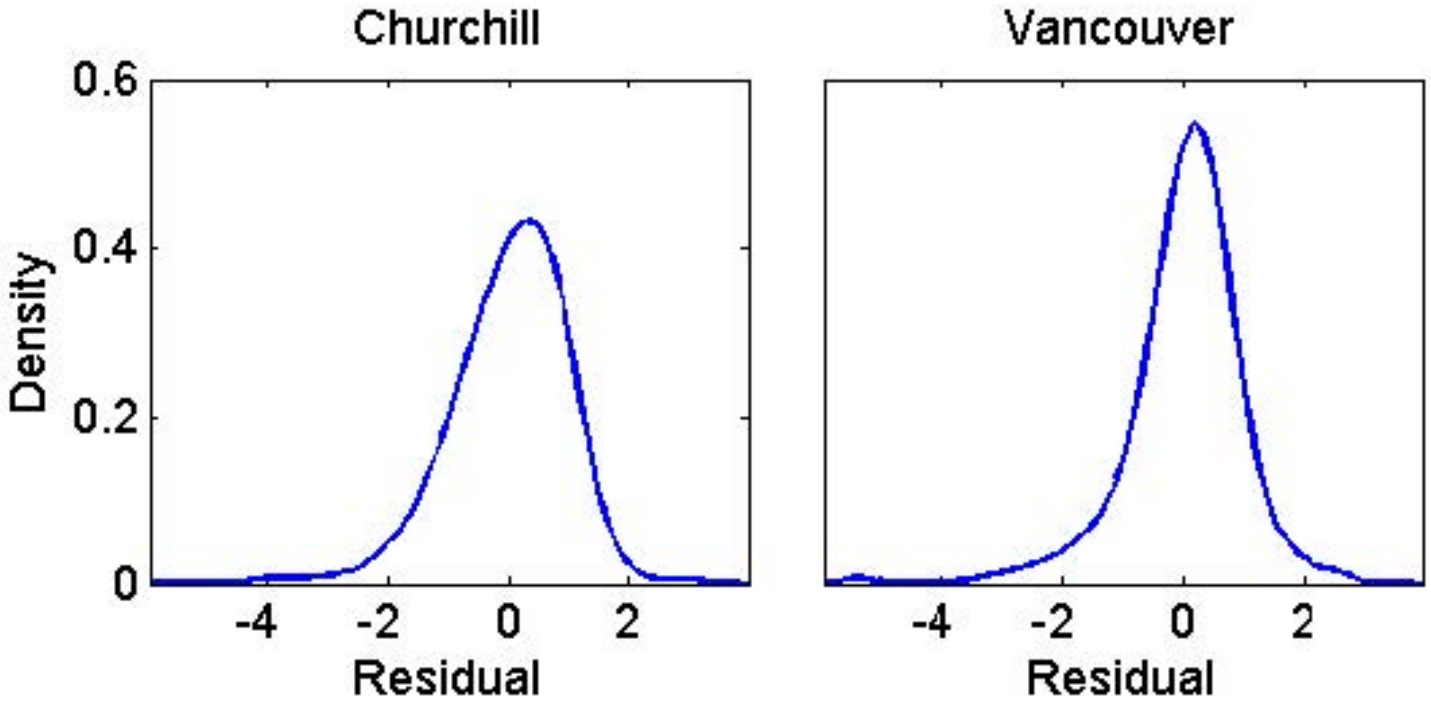
Findings: Precipitation at Prince Rupert

- The sharp peak: most rains come in the form of steady drizzles that leave up to 10 mm of rain.
- Skewness to the right: another main form of rains come as large violent storms that can dump more than 100 mm of rain.

Smooth fit of precipitation data



Estimated density of residuals



- Constrained function can often be expressed as the transformation of an unconstrained function.
- Apply the roughness penalty approach to the unconstrained function:
  - rewrite the loss function in terms of the unconstrained function,
  - use a penalty penalizing the unconstrained function,
  - may introduce additional difficulty in numerical optimization.
- A differential equation expression may provide alternative useful perception.